

Section 13.6: Power Plants and the Electrical Grid

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1. AC power generation was chosen over DC power generation because AC power can work with transformers, which can step up the voltage of AC power and transmit it with little power loss. DC power does not work with transformers.

2. (a) Use the power equation in the form $P = I^2 R$:

$$\begin{aligned} P &= I^2 R \\ &= (2\text{ kA})^2 (10\ \Omega) \\ &= (2000\text{ A})^2 (10\ \Omega) \\ &= 4 \times 10^7\text{ W} \end{aligned}$$

$$P = 40\text{ MW}$$

So 40 MW of power is transformed to unusable thermal energy. The percentage of power lost can now be found:

$$\frac{40\text{ MW}}{200\text{ MW}} \times 100 = 20\%$$

So 20 % of the power was lost.

(b) Use the power equation in the form $P = I^2 R$:

$$\begin{aligned} P &= I^2 R \\ &= (200\text{ A})^2 (10\ \Omega) \\ &= 4 \times 10^5\text{ W} \end{aligned}$$

$$P = 0.4\text{ MW}$$

So 0.4 MW of power is transformed to unusable thermal energy. The percentage of power lost can now be found:

$$\frac{0.4\text{ MW}}{200\text{ MW}} \times 100 = 0.2\%$$

So 0.2 % of the power was lost.

(c) Use the power equation in the form $P = I^2 R$:

$$\begin{aligned} P &= I^2 R \\ &= (3000\text{ A})^2 (0.50\ \Omega) \\ &= 4.5 \times 10^6\text{ W} \end{aligned}$$

$$P = 4.5\text{ MW}$$

So 4.5 MW of power is transformed to unusable thermal energy. The percentage of power lost can now be found:

$$\frac{4.5\text{ MW}}{10\text{ MW}} \times 100 = 45\%$$

So 45 % of the power was lost.

3. Electrical energy cannot be easily stored so generating electrical energy on demand ensures that most of the electrical energy produced is used quickly, which reduces waste.

4. Commercial generators have multiple armatures and coils, use electromagnets, and must be maintained at a desired frequency. The electrical generators described in Section 13.4 did not have these features.

5. The connection point in the middle of the coil is used to create the three-wire system for household circuits that was described in Section 13.3. In the three-wire system, the wire connected to the middle of the transformer is the white wire and the wires connected on the ends are the red and black wires.

6. No, it would not be possible for the generator to power itself. A generator is not 100 % efficient because energy is always converted into unusable thermal energy through the motion of the generator and the resistance in the wires, so a generator powered by itself would have continuously less input energy and would eventually stop.